

Surveying Exam Solutions — Questions 2 & 3

Course: CEE 308 — Engineering Surveying

Institution: Nile University of Nigeria, Faculty of Engineering

Level: 300 Level

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Question 2 (40 marks) — Horizontal Circular Curve Setting-Out

Problem Statement

Two straights projected forward meet at an intersection point I. Given:

Parameter	Value
Deflection angle (Δ)	40°
Radius (R)	300 m
Chord length (C)	25 m
Chainage of I	1786.59 m

Task: Tabulate all setting-out data assuming 25-m chords on a through chainage basis.

Step 1: Compute the Tangent Length (T)

The tangent length is the distance from the intersection point (I) to the beginning of the curve (BC).

Formula:

$$T = R \times \tan(\Delta / 2)$$

Calculation:

$$\begin{aligned}\Delta / 2 &= 40^\circ / 2 = 20^\circ \\ T &= 300 \times \tan(20^\circ) \\ T &= 300 \times 0.363970 \\ T &= 109.19 \text{ m}\end{aligned}$$

Step 2: Compute the Length of Curve (L)

Formula:

$$L = R \times \Delta \text{ (in radians)}$$

Convert Δ to radians:

$$\Delta_{\text{rad}} = 40 \times \pi / 180 = 40 \times 0.0174533 = 0.698132 \text{ rad}$$

$$L = 300 \times 0.698132$$

$$L = 209.44 \text{ m}$$

Step 3: Determine the Chainages of BC and EC

The Beginning of Curve (BC), also called the Point of Curve (PC), is located one tangent length before the intersection point:

$$\begin{aligned} \text{Chainage of BC} &= \text{Chainage of I} - T \\ &= 1786.59 - 109.19 \\ &= 1677.40 \text{ m} \end{aligned}$$

The End of Curve (EC), also called the Point of Tangency (PT), is at:

$$\begin{aligned} \text{Chainage of EC} &= \text{Chainage of BC} + L \\ &= 1677.40 + 209.44 \\ &= 1886.84 \text{ m} \end{aligned}$$

Step 4: Compute the Deflection Angle per Chord

The deflection angle for each full chord of length C is:

Formula:

$$\delta = C / (2R) \quad [\text{in radians}]$$

Calculation:

$$\delta_{\text{rad}} = 25 / (2 \times 300) = 25 / 600 = 0.041667 \text{ rad}$$

Convert to degrees, minutes, seconds:

$$\delta_{\text{deg}} = 0.041667 \times (180 / \pi) = 2.387^\circ = 2^\circ 23' 14.4''$$

This means each 25 m chord corresponds to a deflection of **2° 23' 14.4"** from the tangent at BC.

Step 5: Build the Setting-Out Table

Starting from BC at chainage 1677.40 m, place stations at every 25 m interval. The cumulative deflection at each station is the station number $\times \delta$.

For the last sub-chord (from the last full station to EC), the chord length is:

$$\begin{aligned} \text{Last chord} &= \text{Chainage of EC} - \text{Chainage of last full station} \\ &= 1886.84 - 1877.40 \\ &= 9.44 \text{ m} \end{aligned}$$

The deflection for the sub-chord is proportional:

$$\delta_{\text{sub}} = (9.44 / 25) \times 2^\circ 23' 14.4'' = 0^\circ 54' 5.2''$$

The cumulative deflection at EC must equal $\Delta/2 = 20^\circ 0' 0''$ exactly.

Full Setting-Out Table:

Station	Chainage (m)	Chord (m)	Deflection Angle	Cumulative Deflection
BC	1677.40	—	0° 0' 0"	0° 0' 0"
BC+1	1702.40	25.00	2° 23' 14.4"	2° 23' 14.4"
BC+2	1727.40	25.00	2° 23' 14.4"	4° 46' 28.8"
BC+3	1752.40	25.00	2° 23' 14.4"	7° 9' 43.2"
BC+4	1777.40	25.00	2° 23' 14.4"	9° 32' 57.6"
BC+5	1802.40	25.00	2° 23' 14.4"	11° 56' 12.0"
BC+6	1827.40	25.00	2° 23' 14.4"	14° 19' 26.4"
BC+7	1852.40	25.00	2° 23' 14.4"	16° 42' 40.8"
BC+8	1877.40	25.00	2° 23' 14.4"	19° 5' 55.2"
EC	1886.84	9.44	0° 54' 5.2"	20° 0' 0"

Verification: Total cumulative deflection at EC = $20^{\circ} 0' 0'' = \Delta/2 \checkmark$

Question 3 (40 marks) — Error Propagation and Traverse Adjustment

Part (i): Triangle Area and Standard Error

Given:

Quantity	Symbol	Value	Standard Deviation
Side a	a	165.22 m	$\sigma_a = \pm 0.046 \text{ m}$
Side b	b	68.25 m	$\sigma_b = \pm 0.018 \text{ m}$
Included angle	C	$40^{\circ}30'$	$\sigma_C = \pm 5'15''$

Task: Compute the area of the triangle and the standard error of the area.

Step 1: Convert the Angle to Decimal Degrees and Radians

$$C = 40^{\circ} + 30/60 = 40.5^{\circ}$$

For the standard deviation of the angle, convert arcseconds to radians:

$$\begin{aligned}\sigma_C &= 5 \text{ minutes } 15 \text{ seconds} = 5.25 \text{ minutes} = 5.25/60 \text{ degrees} = 0.0875^{\circ} \\ \sigma_C (\text{radians}) &= 0.0875 \times \pi/180 = 0.001527 \text{ rad}\end{aligned}$$

Step 2: Compute the Area

Formula:

$$A = \frac{1}{2} \times a \times b \times \sin(C)$$

Calculation:

$$\sin(40.5^{\circ}) = 0.649448$$

$$A = 0.5 \times 165.22 \times 68.25 \times 0.649448$$
$$A = 3661.67 \text{ m}^2$$

Result: Area = **3661.67 m²**

Step 3: Compute the Partial Derivatives

For error propagation, we need the partial derivatives of A with respect to each variable:

$$\partial A / \partial a = \frac{1}{2} \times b \times \sin(C) = 0.5 \times 68.25 \times 0.649448 = 22.162 \text{ m}$$

$$\partial A / \partial b = \frac{1}{2} \times a \times \sin(C) = 0.5 \times 165.22 \times 0.649448 = 53.651 \text{ m}$$

$$\begin{aligned} \partial A / \partial C &= \frac{1}{2} \times a \times b \times \cos(C) \\ &= 0.5 \times 165.22 \times 68.25 \times \cos(40.5^\circ) \\ &= 0.5 \times 165.22 \times 68.25 \times 0.760406 \\ &= 4287.27 \text{ m}^2/\text{rad} \end{aligned}$$

Note: $\partial A / \partial C$ must be in **radians** because the error σ_C was converted to radians. The chain rule requires consistent units.

Step 4: Apply the Error Propagation Formula

Formula:

$$\sigma_A^2 = (\partial A / \partial a \times \sigma_a)^2 + (\partial A / \partial b \times \sigma_b)^2 + (\partial A / \partial C \times \sigma_C)^2$$

Calculation:

$$(\partial A / \partial a \times \sigma_a)^2 = (22.162 \times 0.046)^2 = (1.019)^2 = 1.039$$

$$(\partial A / \partial b \times \sigma_b)^2 = (53.651 \times 0.018)^2 = (0.966)^2 = 0.933$$

$$(\partial A / \partial C \times \sigma_C)^2 = (4287.27 \times 0.001527)^2 = (6.547)^2 = 42.858$$

$$\sigma_A^2 = 1.039 + 0.933 + 42.858 = 44.830$$

$$\sigma_A = \sqrt{44.830} = 6.70 \text{ m}^2$$

Step 5: State the Final Result

$$\text{Area} = 3661.67 \pm 6.70 \text{ m}^2$$

The dominant error source is the angle uncertainty (contributing 42.858 out of 44.830 in the variance sum), which makes sense given the relatively large angular error of 5'15".

Part (ii): Bowditch Traverse Adjustment

Given traverse (closed loop ABCDEF):

Line	Length (m)	WCB	ΔE (m)	ΔN (m)
AB	520.5	<i>derived</i>	0.000	-450.500
BC	940.5	<i>derived</i>	745.400	-334.300
CD	516.0	<i>derived</i>	-123.000	532.200
DE	618.8	<i>derived</i>	498.700	426.900
EF	622.5	293°39'33"	?	?
FA	406.2	284°42'01"	?	?

The ΔE and ΔN for EF and FA are not given — we compute them from the WCB and length.

Step 1: Compute ΔE and ΔN for EF and FA

For a line of length L and whole circle bearing θ :

$$\Delta E = L \times \sin(\theta)$$

$$\Delta N = L \times \cos(\theta)$$

For EF ($L = 622.5$ m, $\theta = 293^\circ 39' 33'' = 293.6592^\circ$):

$$\Delta E_{EF} = 622.5 \times \sin(293.6592^\circ) = 622.5 \times (-0.91616) = -570.178 \text{ m}$$

$$\Delta N_{EF} = 622.5 \times \cos(293.6592^\circ) = 622.5 \times 0.40036 = 249.806 \text{ m}$$

For FA ($L = 406.2$ m, $\theta = 284^\circ 42' 01'' = 284.7003^\circ$):

$$\Delta E_{FA} = 406.2 \times \sin(284.7003^\circ) = 406.2 \times (-0.96719) = -392.904 \text{ m}$$

$$\Delta N_{FA} = 406.2 \times \cos(284.7003^\circ) = 406.2 \times 0.25404 = 103.078 \text{ m}$$

Step 2: Check Traverse Closure (Misclosure)

For a closed loop, $\Sigma\Delta E$ and $\Sigma\Delta N$ should both be zero.

$$\begin{aligned}\Sigma\Delta E &= 0.000 + 745.400 + (-123.000) + 498.700 + (-570.178) + (-392.904) \\ &= 158.018 \text{ m}\end{aligned}$$

$$\begin{aligned}\Sigma\Delta N &= (-450.500) + (-334.300) + 532.200 + 426.900 + 249.806 + 103.078 \\ &= 527.185 \text{ m}\end{aligned}$$

Misclosure vector: (158.018, 527.185) m

Misclosure magnitude: $\sqrt{(158.018^2 + 527.185^2)} = 550.36 \text{ m}$

Perimeter: $\Sigma L = 520.5 + 940.5 + 516.0 + 618.8 + 622.5 + 406.2 = 3624.5 \text{ m}$

The large misclosure indicates significant observational errors that need adjustment.

Step 3: Apply Bowditch (Compass Rule) Corrections

Principle: The Bowditch method distributes the misclosure in proportion to the length of each line. The correction to each ΔE and ΔN is:

$$\begin{aligned}\text{Correction to } \Delta E_i &= -\Sigma\Delta E \times (L_i / \Sigma L) \\ \text{Correction to } \Delta N_i &= -\Sigma\Delta N \times (L_i / \Sigma L)\end{aligned}$$

The negative sign is because we apply corrections opposite to the misclosure to close the loop.

Bowditch Correction Factors:

Line	Length (m)	ΔE Correction	ΔN Correction
AB	520.5	$-158.018 \times 520.5/3624.5 =$ -22.692	$-527.185 \times 520.5/3624.5 =$ -75.707
BC	940.5	$-158.018 \times 940.5/3624.5 =$ -41.003	$-527.185 \times 940.5/3624.5 =$ -136.796
CD	516.0	$-158.018 \times 516.0/3624.5 =$ -22.496	$-527.185 \times 516.0/3624.5 =$ -75.052
DE	618.8	$-158.018 \times 618.8/3624.5 =$ -26.978	$-527.185 \times 618.8/3624.5 =$ -90.005
EF	622.5	$-158.018 \times 622.5/3624.5 =$ -27.139	$-527.185 \times 622.5/3624.5 =$ -90.543

Line	Length (m)	ΔE Correction	ΔN Correction
FA	406.2	$-158.018 \times 406.2/3624.5 =$ -17.709	$-527.185 \times 406.2/3624.5 =$ -59.082

Adjusted ΔE and ΔN :

Line	Orig ΔE	Corr	Adj ΔE	Orig ΔN	Corr	Adj ΔN
AB	0.000	-22.692	-22.692	-450.500	-75.707	-526.207
BC	745.400	-41.003	704.397	-334.300	-136.796	-471.096
CD	-123.000	-22.496	-145.496	532.200	-75.052	457.148
DE	498.700	-26.978	471.722	426.900	-90.005	336.895
EF	-570.178	-27.139	-597.317	249.806	-90.543	159.263
FA	-392.904	-17.709	-410.613	103.078	-59.082	43.996

Check: $\Sigma \text{Adj } \Delta E = -22.692 + 704.397 - 145.496 + 471.722 - 597.317 - 410.613 \approx 0 \checkmark$

Check: $\Sigma \text{Adj } \Delta N = -526.207 - 471.096 + 457.148 + 336.895 + 159.263 + 43.996 \approx 0 \checkmark$

Step 4: Compute Adjusted Coordinates

Start at A = (0, 0). Accumulate adjusted ΔE and ΔN sequentially.

A = (0.000, 0.000)	
B = A + (Adj ΔE_{AB} , Adj ΔN_{AB})	
= (0.000 + (-22.692), 0.000 + (-526.207))	
= (-22.692, -526.207)	
C = B + (Adj ΔE_{BC} , Adj ΔN_{BC})	
= (-22.692 + 704.397, -526.207 + (-471.096))	
= (681.705, -997.303)	
D = C + (Adj ΔE_{CD} , Adj ΔN_{CD})	
= (681.705 + (-145.496), -997.303 + 457.148)	
= (536.209, -540.155)	
E = D + (Adj ΔE_{DE} , Adj ΔN_{DE})	
= (536.209 + 471.722, -540.155 + 336.895)	
= (1007.931, -203.260)	
F = E + (Adj ΔE_{EF} , Adj ΔN_{EF})	
= (1007.931 + (-597.317), -203.260 + 159.263)	

$$= (410.614, -43.997)$$

$$\begin{aligned} A_{\text{check}} &= F + (\text{Adj } \Delta E_{\text{FA}}, \text{Adj } \Delta N_{\text{FA}}) \\ &= (410.614 + (-410.613), -43.997 + 43.996) \\ &= (0.001, -0.001) \approx (0, 0) \checkmark \end{aligned}$$

Final Adjusted Coordinates:

Point	Easting (m)	Northing (m)
A	0.000	0.000
B	-22.692	-526.207
C	681.705	-997.303
D	536.209	-540.155
E	1007.931	-203.260
F	410.614	-43.997

Step 5: Compute Lengths and Bearings of Required Lines

For each line from point P_1 to P_2 :

$$\begin{aligned} \text{Length} &= \sqrt{(E_2 - E_1)^2 + (N_2 - N_1)^2} \\ \text{Bearing (WCB)} &= \text{atan2}(E_2 - E_1, N_2 - N_1) \quad [\text{converted to } 0^\circ\text{--}360^\circ] \end{aligned}$$

AD: From A(0, 0) to D(536.209, -540.155)

$$\begin{aligned} \Delta E &= 536.209 - 0 = 536.209 \text{ m} \\ \Delta N &= -540.155 - 0 = -540.155 \text{ m} \\ \text{Length} &= \sqrt{(536.209^2 + 540.155^2)} = 761.11 \text{ m} \\ \text{Bearing} &= \text{atan2}(536.209, -540.155) \\ \text{WCB} &= 135^\circ 12' 36.4'' \end{aligned}$$

BE: From B(-22.692, -526.207) to E(1007.931, -203.260)

$$\begin{aligned} \Delta E &= 1007.931 - (-22.692) = 1030.623 \text{ m} \\ \Delta N &= -203.260 - (-526.207) = 322.947 \text{ m} \\ \text{Length} &= \sqrt{(1030.623^2 + 322.947^2)} = 1080.04 \text{ m} \end{aligned}$$

Bearing = $\text{atan2}(1030.623, 322.947)$
WCB = $72^\circ 36' 05.6''$

BD: From B(-22.692, -526.207) to D(536.209, -540.155)

$\Delta E = 536.209 - (-22.692) = 558.901 \text{ m}$
 $\Delta N = -540.155 - (-526.207) = -13.948 \text{ m}$

Length = $\sqrt{(558.901^2 + 13.948^2)} = 559.07 \text{ m}$

Bearing = $\text{atan2}(558.901, -13.948)$
WCB = $91^\circ 25' 46.6''$

CF: From C(681.705, -997.303) to F(410.614, -43.997)

$\Delta E = 410.614 - 681.705 = -271.091 \text{ m}$
 $\Delta N = -43.997 - (-997.303) = 953.306 \text{ m}$

Length = $\sqrt{(271.091^2 + 953.306^2)} = 991.10 \text{ m}$

Bearing = $\text{atan2}(-271.091, 953.306)$
WCB = $344^\circ 07' 33.1''$

DF: From D(536.209, -540.155) to F(410.614, -43.997)

$\Delta E = 410.614 - 536.209 = -125.595 \text{ m}$
 $\Delta N = -43.997 - (-540.155) = 496.158 \text{ m}$

Length = $\sqrt{(125.595^2 + 496.158^2)} = 511.81 \text{ m}$

Bearing = $\text{atan2}(-125.595, 496.158)$
WCB = $345^\circ 47' 41.3''$

Summary of Results

Question 2 — Curve Setting-Out:

Quantity	Value
Tangent length T	109.19 m
Curve length L	209.44 m
Chainage of BC	1677.40 m
Chainage of EC	1886.84 m

Quantity	Value
Deflection per 25 m chord	2° 23' 14.4"

Question 3(i) — Triangle:

Quantity	Value
Area	3661.67 m ²
Standard error	±6.70 m ²

Question 3(ii) — Traverse Coordinates:

Point	E (m)	N (m)
A	0.000	0.000
B	-22.692	-526.207
C	681.705	-997.303
D	536.209	-540.155
E	1007.931	-203.260
F	410.614	-43.997

Question 3(ii) — Required Lines:

Line	Length (m)	Bearing (WCB)
AD	761.11	135° 12' 36.4"
BE	1080.04	72° 36' 05.6"
BD	559.07	91° 25' 46.6"
CF	991.10	344° 07' 33.1"
DF	511.81	345° 47' 41.3"